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ABSTRACT

A paleomagnetic analysis of units of Devonian to Triassic age within the Redding section of the eastern Klamath Mountains indicates clockwise rotation of 83°–116° with respect to stable North America. In the kinematic model presented here, shear zones within the Franciscan Complex, with separation antithetic to the San Andreas, are used to effect this rotation.

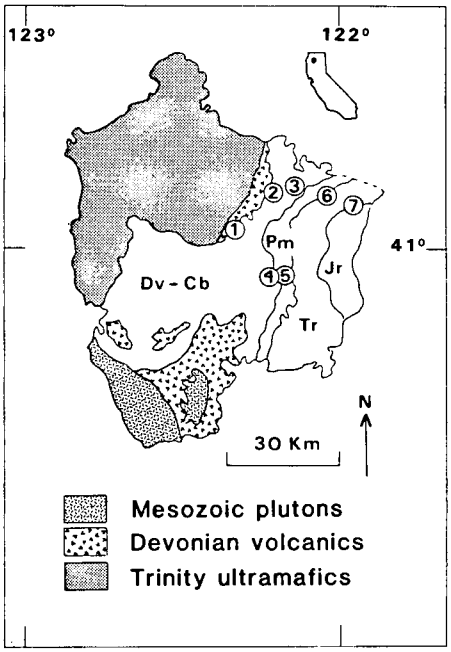


Figure 1. Geology of Redding section showing seven sampling sites. Numbers are keyed to data in Table 1.

INTRODUCTION

In the past ten years several paleomagnetic studies reported a declination anomaly in the Pacific northwest of the United States. The data suggest that a mobile terrane, consisting of the Oregon Coast Range, Cascades, Blue Mountains, and perhaps the Klamath Mountains, has rotated clockwise as much as 80° (Simpson and Cox, 1977; Wilson and Cox, 1980; Hillhouse and Gromme, 1980). The progressive increase in rotation with age in the Oregon Coast Range and the Cascades indicates that much of the rotation is a Cenozoic event. We present here additional paleomagnetic data from the eastern Klamath belt, northern California, and a kinematic model to account for rotational movements.

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SAMPLING AND ANALYSIS

Of all the Klamath terranes, the Redding section of the eastern Klamath belt is uniquely suited for paleomagnetic study. It contains a continuous stratigraphic succession, extending from Devonian to Jurassic, that forms a well-understood structure consisting of a flat western segment and an eastwardly dipping eastern segment (Fig. 1). The eastern homoclinal segment forms a gentle flexure, changing from a northern to a northeastern strike.

Ninety-two samples were collected at seven outcrops, ranging in age from Devonian to Late Triassic (see Fig. 1 and Table 1). For six weeks prior to the measurements, the samples were stored in a magnetically shielded room to reduce possible viscous magnetization. The clastic rocks (Kennett and Bragdon Formations, undifferentiated Triassic) were thermally demagnetized in 100° increments from 200° to 600°, and the carbonate rocks (McCloud and Modin Formations) in 50° steps from 200° to 400°; the volcanic rocks (Nosoni Formation) were progressively demagnetized by alternating fields up to 40 mT. The samples remained in the shielded room throughout the experiment.

The directions of magnetization were analyzed by calculating linear trends in orthogonal vector projections (As-

TABLE 1. RESULTS OF PALEOMAGNETIC ANALYSIS*

	Formation	Age	N	R	D	I	K	α	Lat.	Long.	dp	dm	ΔD	Δlat
1.	Kennett	Middle Devonian	6/10	5.25 5.25	77.1 67.2	29.0 52.0	6.6 6.6	28.1 28.1	19.7 36.9	323.8 313.5	17.1 26.3	31.0 38.4	115.6	1.7N
2.	Lower Bragdon	Carboniferous	10/18	9.20 9.20	52.3 3.2	35.1 66.4	11.2 11.2	15.1 15.1	40.8 81.7	337.1 252.5	10.0 20.3	17.4 24.8	95.2	7.0N
4.	McCloud	Permian	10/12	9.85 9.85	228.9 150.9	-29.3 -82.0	61.0 60.4	6.2 6.3	41.0 53.9	343.7 224.7	3.8 11.5	6.9 12.2	90.9	5.5N
5.	Nosoni	Permian	11/13	9.57 9.61	149.4 349.9	62.2 82.3	7.0 7.2	18.6 18.3	- 1.4 55.6	259.3 233.0	22.5 34.7	28.9 35.6	-	-
7.	Modin	Triassic	13/16	12.72 12.72	70.1 26.3	23.6 44.4	43.6 43.6	6.3 6.3	23.1 63.8	331.1 353.5	3.6 5.0	6.8 8.0	82.9	11.1S

*Upper values at each site are structurally corrected data; lower values structurally uncorrected.

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Zijderveld diagrams), examples of which are shown in Figure 2. All samples from the upper Bragdon Formation and the Triassic clastics behaved erratically, and both sites were rejected. The remaining five sites were analyzed in two steps. The Fisher mean was calculated for each site. Samples whose directions deviated from the mean by more than three times $\alpha 95$ were then deleted and a new mean was calculated. The latter statistics are listed in Table 1, and the pole positions are shown in Figure 3.

RESULTS

All pole positions differ significantly from the time-equivalent data for stable North America (Irving, 1979), and it is important to establish whether the rocks retained their primary magnetization. The basalts from the Permian Nosoni Formation have clearly been remagnetized. Not only is their pole position vastly different from the pole positions of the other sites, but the different flow units record both normal and reverse polarities that are incompatible with a Permian magnetization.

We believe that the remaining four sites have preserved their primary magnetization for the following reasons. (1) Their structurally corrected pole positions are displaced in a uniform way from the North American polar wander path. The structurally uncorrected poles, on the other hand, scatter, and with the exception of the result from the Lower Bragdon Formation, their ellipses of confidence do not overlap with any part of the Mesozoic and Cenozoic polar wander path of stable North America. (2) The paleolatitudes for the structurally corrected sites are in good agreement with latitudes predicted by the stable North America polar wandering curve given a primary magnetization (Fig. 4). If magnetization was induced after tilting, this would be a fortuitous coincidence. In contrast, the structurally uncorrected data vary in an unsystematic way from the stable North American curve, regardless of the age of magnetization chosen. (3) The Lower Permian McCloud data are reversed, whereas the Upper Triassic Modin data are normal. This polarity agrees with that predicted, given the age of the units, and argues for a primary magnetization.

Recently, Achache et al. (1982) reported paleomagnetic results from the Kennett limestone. A fold test proves that their samples have been remagnetized, and they argued that the remagnetization was caused by the nearby Shasta Bally batholith at 136 m.y. during a reversed interval of 1 m.y. duration. Our sampling site lies 30 km farther north and should not have been affected by this intrusion. Indeed, our samples are of opposite polarity, and the direction of magnetization that we obtained is very different from Achache et al.'s value (D = 218.7, I = 49.4), whether or not we apply the structural correction to our data.

The last two columns in Table 1 list the difference in declination and latitude between our data and the data for stable North America (Irving, 1979). If primary magnetization has been preserved, then the results indicate that the Redding section has rotated clockwise between 83° and 116° with respect to stable North America (Table 1). Figure 5 shows that most of the movement occurred after Late Triassic time but suggests that additional Mesozoic and Paleozoic movement may have occurred. The latitude changes do not show a systematic trend and, within the error limits, do not indicate any north-south displacement of the Klamath terrane.

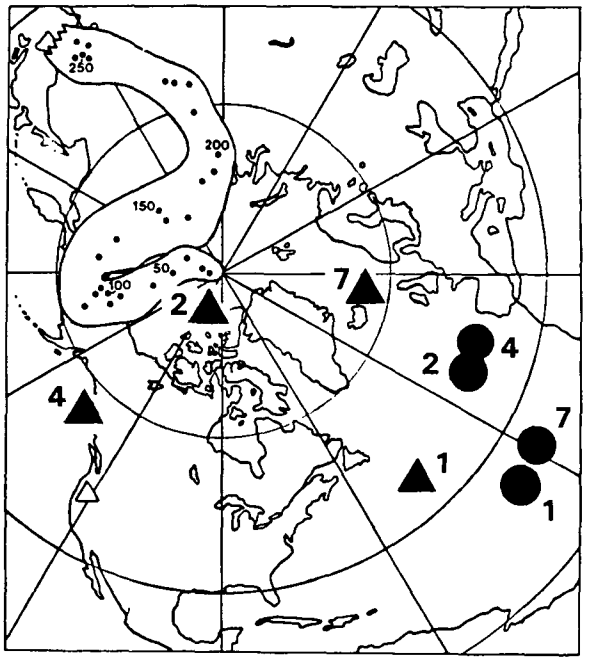


Figure 3. Pole positions for structurally corrected (circles) and uncorrected (triangles) sites 1, 2, 4, and 7. Open triangle is sampling site. Polar wandering curve for cratonic North America from Irving (1979).

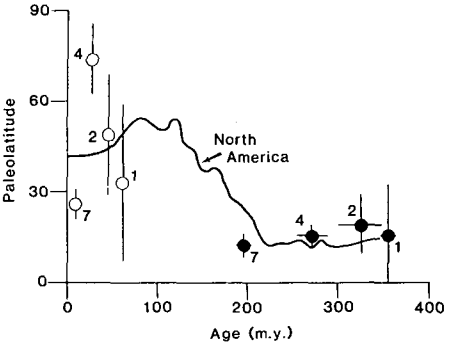


Figure 4. Paleolatitudes of Redding section units for structurally corrected (solid circles) and uncorrected (open circles) data. Age position shown for uncorrected data is arbitrary. Vertical and horizontal bars indicate uncertainty in latitude and ages. Heavy black line is paleolatitude of Redding section if it were coupled to stable North America. Data from Irving (1979).

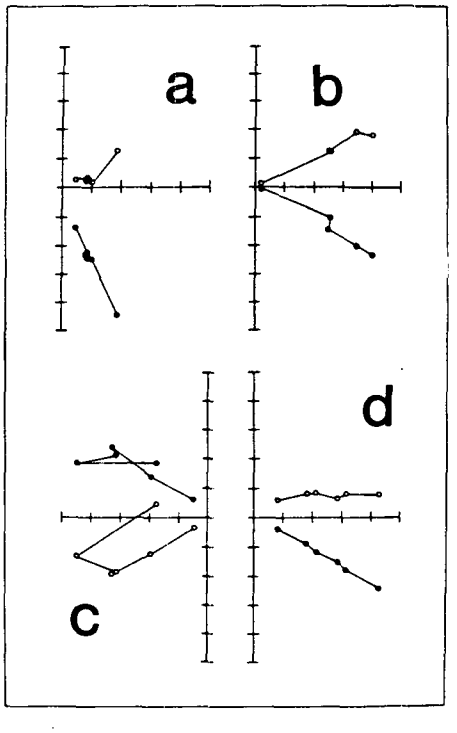


Figure 2. As-Zijderveld diagrams showing vector trends with demagnetization for four samples. Vector is projected onto horizontal plane (open circles) and vertical, east-west–striking plane (solid circles); a = Kennett, b = Bragdon, c = McCloud, d = Modin.

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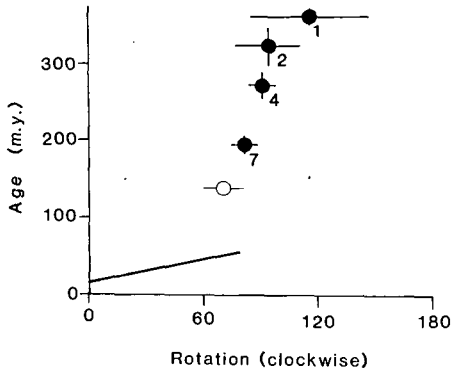


Figure 5. Plot of rotational displacement of Redding section relative to stable North America. Horizontal and vertical lines depict uncertainty in declination and ages of sampled units. Open circle represents result of Achache et al. (1982). Heavy black line indicates well-documented progressive rotation of Oregon Coast Range and Cascade provinces (Beck and Plumley, 1980).

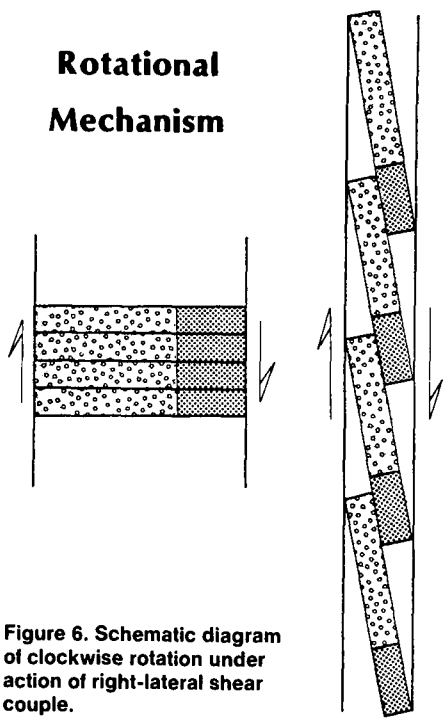


Figure 6. Schematic diagram of clockwise rotation under action of right-lateral shear couple.

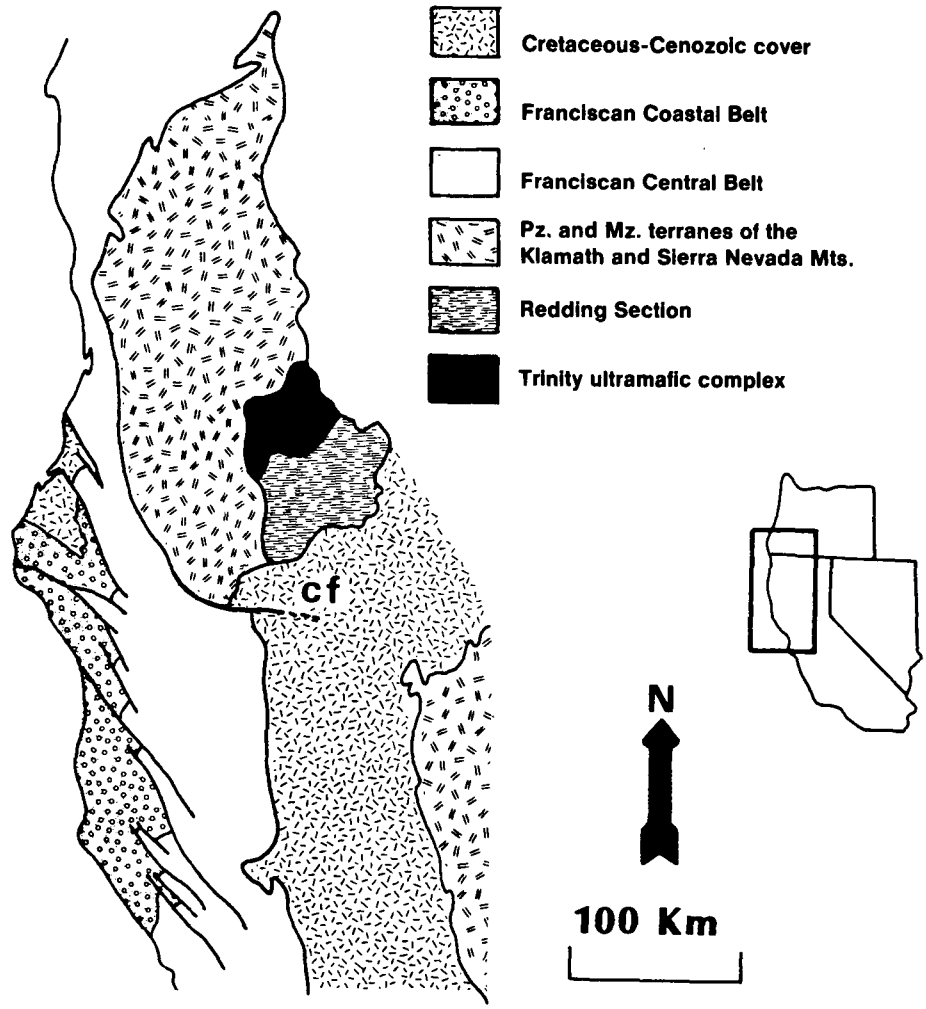


Figure 7. Geology of northern California. Note series of northwest-trending faults displaying left-lateral offset of Coastal Belt–central Franciscan contact. CF = Cold Fork fault zone.

KINEMATIC MODEL

The model we present assumes that (1) the Klamath rotations occurred in the Cenozoic in conjunction with contemporaneous movements in the Oregon Coast Range and Cascades, and that (2) the rotations resulted from right-lateral shear acting on the western margin of North America since 60 m.y. ago (Atwater and Molnar, 1973; Jackson et al., 1978). Admittedly, the Redding section rotation has been shown only to be post–Late Triassic and could therefore have occurred prior to the Cenozoic (perhaps during an accretion event). However, we are impressed by the similarity in the amount of rotation in the Oregon Coast Range (Eocene) and the youngest Redding section sites, and so we prefer to connect the two.

The manner in which rotations occur within a wrench zone is illustrated in the clay-cake experiments of Cloos (1955) and Wilcox et al. (1973). It involves movement along faults antithetic to the main wrench fault and a narrowing and lengthening of the wrench zone (Fig. 6). This mechanism has been suggested to explain rotations in the Middle East (Freund and Tarling, 1979), southern California (Garfunkel, 1974; Luyendyk et al., 1980; Terres and Sylvester, 1981), and Iran (Freund, 1970).

Figure 6 is a schematic of this rotational mechanism exaggerated to create 70° of clockwise rotation. Movement occurs along faults on which offset is antithetic to the main right-lateral shear system. The strains resulting from this rotation are large; elongation and shortening are 5.20 and 0.70 in Figure 6. Considerable gaps are created, causing the area of the shear zone to be nearly doubled. Consequently, the structural features predicted by this model are a series of faults subparallel to the right-lateral Pacific margin fault system that display *left-lateral*, antithetic separation. Grabens or sedimentary basins may be expected to appear at intervals along the plate margin.

Although these structural features are not apparent within the Klamath Mountains, they are well expressed in the underlying Franciscan to the southwest, where several northwest-trending faults offset the contact between the Coastal and Central Franciscan belts (Fig. 7). These faults are subparallel to the San Andreas and display antithetic, left-lateral separation. A consideration of the geometry of Figure 6 reveals that for each slab, the ratio of thickness to offset is equal to the tangent of the angle of rotation. Applying this formulation to the Franciscan, a rotation of about 70° is obtained.

Kleist (1974) and Kramer (1976) both mapped these offsets, and neither has interpreted them in this fashion. Both view the contact as having been folded under the influence of right-lateral movements. How-

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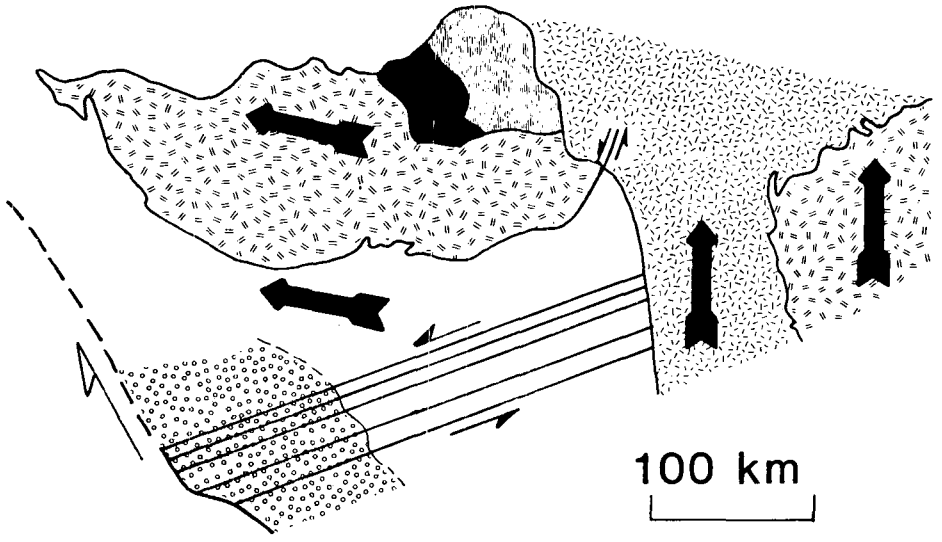


Figure 8. Pre-Cenozoic reconstruction of northern California. Rotation is facilitated by mechanism of Figure 4 (70°) and by movement along Cold Fork fault zone (9°). Arrows show direction of present north in each terrane.

ever, the field evidence for this is not compelling due to the lack of exposure of the contact.

The grabens predicted by Figure 6 may be represented by several basins situated along the Pacific coast of California (Silver, 1971; Hoskins and Griffiths, 1971). Prominent among these is the Eel River Basin, in which wells have penetrated more than 2,680 m (8,800 ft) of Miocene and younger sediments (Ogle, 1953).

An additional rotation mechanism is suggested by the curved outline of the Klamath arc. At the southernmost point of the arc, Lower Cretaceous Great Valley sedimentary units are separated left-laterally along the Cold Fork fault zone (CF in Fig. 7). This occurs at the point where underlying basement changes from Franciscan to Klamath (Worrall, 1979). It suggests that movement along the arcuate fault that separates uppermost Franciscan (South Fork Mountain Schist) from lowermost Klamath (Galice) might account for some of the Klamath rotation. The Great Valley separation measures about 30 km; the radius of the Klamath arc is about 200 km. Therefore, an additional 9° of clockwise rotation could have occurred by movement between the Klamath and Franciscan terranes. Geologic evidence suggests that this movement occurred in the Cretaceous (Jones and Irwin, 1971), and may have preceded the motion along the Franciscan faults.

Figure 8 is a restoration of the Klamath Mountains to its pre-Cenozoic orientation. The rotation is accomplished by regarding the Klamaths as coupled to rotating Franciscan basement (70° of rotation) and to a subordinate extent by rotation along the arcuate Cold Fork fault zone (9° of rotation). Note that the entire Klamath province is regarded as a rigid body that remains internally undeformed and coupled to underlying and rotating Franciscan.

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